GLOBAL STUDY SUPPORT / REFERENCE SERIES

Low-Rise Frame Concept Study

ILLUSTRATIVE DEMONSTRATION - NOT CLIENT WORK

Objective

Understand how assumptions connect geometry, loading and structural behaviour in an invented one-bay frame. This is an introductory reference, not a construction design or certified analysis.

Illustrative geometry

Assume one span of 5 m and a storey height of 3 m. Use consistent units. Define connectivity and supports before introducing member properties. No material grade or design code is selected in this conceptual example.

Simple load illustration

For a simply supported 5 m beam under a uniform 10 kN/m load, the idealised reactions are wL/2 = 25 kN each. The maximum bending moment is wL squared/8 = 31.25 kNm. These idealised expressions do not establish member adequacy.

Review before analysis

Confirm load path, end releases, support restraints, member orientation, diaphragm assumptions and self-weight treatment. Check equilibrium and deformation qualitatively before accepting a software output.

Available deliverable and limits

This PDF and the study notes are the available deliverables. No Revit or ETABS model, code compliance assessment or reinforcement design is included. Actual design needs site data, applicable standards and qualified professional verification.